

SECHELT, BC, Canada

WMO: 716380

Lat: 49.4581N

Lon: 123.7153W

Elev: 282

StdP: 14.55

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>12     | (b)<br>24.3 | (c)<br>26.6 | (d)<br>11.3                   | (e)<br>9.9 | (f)<br>28.9 | (g)<br>18.0 | (h)<br>13.8 | (i)<br>30.1 | (j)<br>20.8           | (k)<br>46.7 | (l)<br>19.2 | (m)<br>46.3 | (n)<br>3.6            | (o)<br>330 | (p)<br>0.499 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.2                   | 81.2            | 63.3 | 77.5 | 62.3 | 74.2 | 61.5 | 65.2                | 76.8 | 63.8 | 74.4 | 62.6 | 72.0 | 3.5                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 60.8                            | 80.4 | 69.3 | 59.5 | 76.7 | 68.2 | 58.4 | 73.7 | 67.0 | 30.2          | 76.3 | 29.2 | 74.8 | 28.3 | 72.1 | 71.1              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
| Min               |      | Max  |                            | Min  | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 17.2              | 14.4 | 12.3 |                            | DB   | 19.1 | 88.6               | 2.9                                                | 3.3       | 17.0 | 91.0       | 15.3 | 92.9       | 13.7 | 94.7       | 11.6 | 97.1 |
|                   |      |      |                            | WB   | 17.4 | 68.0               | 3.3                                                | 2.0       | 15.0 | 69.4       | 13.1 | 70.6       | 11.3 | 71.7       | 8.9  | 73.1 |

Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 50.0        | 39.0  | 39.2  | 43.0  | 47.4  | 54.6  | 59.0  | 63.9  | 64.0  | 58.2  | 49.6  | 42.9  | 37.9  | (6)  | (7)  |
|                                                                        | DBStd       | 10.55       | 5.41  | 5.50  | 4.72  | 4.14  | 5.00  | 4.73  | 4.11  | 3.99  | 4.66  | 4.29  | 5.94  | 5.58  | (7)  | (8)  |
|                                                                        | HDD50       | 1637        | 343   | 302   | 221   | 98    | 14    | 0     | 0     | 0     | 1     | 59    | 222   | 377   | (8)  | (9)  |
|                                                                        | HDD65       | 5584        | 807   | 722   | 682   | 527   | 323   | 193   | 69    | 68    | 211   | 477   | 663   | 842   | (9)  | (10) |
|                                                                        | CDD50       | 1619        | 0     | 1     | 4     | 21    | 157   | 269   | 431   | 433   | 246   | 47    | 9     | 1     | (10) | (11) |
|                                                                        | CDD65       | 90          | 0     | 0     | 0     | 0     | 0     | 12    | 35    | 37    | 6     | 0     | 0     | 0     | (11) | (12) |
|                                                                        | CDH74       | 775         | 0     | 0     | 0     | 2     | 22    | 121   | 303   | 273   | 54    | 0     | 0     | 0     | (12) | (13) |
|                                                                        | CDH80       | 157         | 0     | 0     | 0     | 0     | 0     | 26    | 70    | 53    | 8     | 0     | 0     | 0     | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 4.7         | 5.1   | 4.7   | 5.7   | 5.3   | 4.3   | 4.5   | 4.0   | 4.0   | 4.1   | 4.6   | 5.3   | 5.2   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 61.6        | 8.7   | 5.9   | 5.6   | 3.9   | 3.0   | 2.5   | 1.7   | 2.0   | 2.7   | 6.8   | 9.3   | 9.0   | (15) | (16) |
|                                                                        | PrecMax     | 72.8        | 15.0  | 10.8  | 9.6   | 7.4   | 4.6   | 4.6   | 3.1   | 7.9   | 6.5   | 11.6  | 16.0  | 11.5  | (16) | (17) |
|                                                                        | PrecMin     | 49.8        | 5.7   | 1.9   | 2.2   | 1.6   | 1.0   | 0.9   | 0.4   | 0.4   | 0.8   | 2.3   | 4.9   | 6.3   | (17) | (18) |
|                                                                        | PrecStd     | 6.4         | 2.4   | 1.9   | 1.5   | 1.5   | 0.9   | 1.0   | 0.7   | 1.8   | 1.5   | 2.3   | 2.9   | 1.5   | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 53.8  | 53.6  | 60.8  | 69.2  | 76.9  | 83.6  | 86.7  | 85.6  | 79.9  | 65.6  | 59.6  | 54.2  | (19) | (20) |
|                                                                        |             | MCWB        | 49.6  | 48.3  | 50.2  | 53.9  | 57.9  | 62.4  | 65.7  | 63.9  | 62.6  | 55.6  | 54.0  | 49.5  | (20) | (21) |
|                                                                        | 2%          | DB          | 51.2  | 51.2  | 56.1  | 61.9  | 72.3  | 77.5  | 81.0  | 80.7  | 74.5  | 61.8  | 55.9  | 50.2  | (21) | (22) |
|                                                                        |             | MCWB        | 47.4  | 46.8  | 48.2  | 50.6  | 57.2  | 61.0  | 63.6  | 64.0  | 60.6  | 55.1  | 52.4  | 46.9  | (22) | (23) |
|                                                                        | 5%          | DB          | 49.2  | 49.1  | 52.9  | 57.9  | 68.2  | 71.8  | 76.9  | 76.7  | 70.4  | 59.0  | 53.4  | 47.9  | (23) | (24) |
|                                                                        |             | MCWB        | 46.1  | 45.2  | 46.8  | 48.9  | 55.9  | 59.4  | 62.8  | 63.3  | 59.6  | 53.8  | 50.2  | 45.3  | (24) | (25) |
|                                                                        | 10%         | DB          | 47.1  | 47.2  | 50.6  | 54.9  | 64.3  | 67.6  | 73.5  | 73.1  | 66.6  | 56.9  | 51.4  | 46.1  | (25) | (26) |
|                                                                        |             | MCWB        | 44.5  | 43.9  | 46.0  | 47.8  | 54.8  | 57.8  | 61.6  | 62.3  | 58.3  | 52.6  | 48.5  | 43.8  | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 50.7  | 50.8  | 51.8  | 54.9  | 60.5  | 65.1  | 68.1  | 66.9  | 64.7  | 58.7  | 55.7  | 51.0  | (27) | (28) |
|                                                                        |             | MCDB        | 53.0  | 52.8  | 58.0  | 65.0  | 71.8  | 77.8  | 82.1  | 79.1  | 74.8  | 62.2  | 57.8  | 53.2  | (28) | (29) |
|                                                                        | 2%          | WB          | 48.4  | 47.8  | 49.3  | 52.5  | 58.7  | 62.7  | 65.2  | 65.3  | 62.4  | 56.6  | 52.8  | 47.5  | (29) | (30) |
|                                                                        |             | MCDB        | 50.4  | 49.7  | 54.1  | 59.6  | 68.4  | 75.1  | 77.3  | 76.4  | 70.6  | 60.0  | 55.6  | 49.5  | (30) | (31) |
|                                                                        | 5%          | WB          | 46.3  | 45.7  | 47.7  | 50.5  | 57.2  | 60.3  | 63.6  | 64.0  | 60.8  | 54.6  | 50.7  | 45.7  | (31) | (32) |
|                                                                        |             | MCDB        | 48.5  | 48.0  | 51.8  | 55.9  | 65.9  | 69.5  | 74.4  | 74.3  | 68.2  | 57.9  | 53.1  | 47.6  | (32) | (33) |
|                                                                        | 10%         | WB          | 44.6  | 44.1  | 46.3  | 48.8  | 55.6  | 58.6  | 62.3  | 62.8  | 59.1  | 52.7  | 48.9  | 43.9  | (33) | (34) |
|                                                                        |             | MCDB        | 47.1  | 46.8  | 50.1  | 53.5  | 63.2  | 66.3  | 71.9  | 71.8  | 64.9  | 56.2  | 51.1  | 45.8  | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.0   | 10.7  | 12.2  | 14.1  | 15.8  | 14.7  | 17.2  | 16.7  | 15.0  | 12.1  | 10.3  | 8.6   | (35) | (36) |
|                                                                        |             | MCDBR       | 10.3  | 12.0  | 15.7  | 19.4  | 23.5  | 23.5  | 24.6  | 24.1  | 22.2  | 15.2  | 10.3  | 9.0   | (36) | (37) |
|                                                                        | 5% WB       | MCWBR       | 7.3   | 8.2   | 9.5   | 10.7  | 11.4  | 10.4  | 10.6  | 11.1  | 11.5  | 9.7   | 7.4   | 6.7   | (37) | (38) |
|                                                                        |             | MCWBR       | 9.5   | 10.0  | 13.8  | 16.0  | 19.7  | 20.0  | 21.9  | 21.0  | 18.2  | 12.8  | 10.0  | 8.8   | (38) | (39) |
| (39) Clear-Sky Solar Irradiance                                        | taub        |             | 0.313 | 0.305 | 0.329 | 0.358 | 0.374 | 0.369 | 0.356 | 0.356 | 0.337 | 0.330 | 0.313 | 0.301 | (39) | (40) |
|                                                                        |             | taud        | 2.444 | 2.518 | 2.466 | 2.414 | 2.405 | 2.432 | 2.503 | 2.509 | 2.563 | 2.557 | 2.528 | 2.467 | (40) | (41) |
|                                                                        | Ebn at Noon |             | 235   | 271   | 281   | 282   | 281   | 283   | 285   | 280   | 276   | 258   | 236   | 224   | (41) | (42) |
|                                                                        |             | Edh at Noon | 20    | 23    | 29    | 34    | 36    | 35    | 32    | 30    | 26    | 22    | 18    | 17    | (42) | (43) |
| (43) All-Sky Solar Radiation                                           | RadAvg      |             | 264   | 537   | 793   | 1244  | 1637  | 1749  | 1962  | 1671  | 1152  | 629   | 310   | 209   | (43) | (44) |
|                                                                        | RadStd      |             | 45    | 85    | 114   | 137   | 153   | 219   | 189   | 140   | 105   | 95    | 61    | 33    | (44) | (45) |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (62 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |